

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 40

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark.</i> • Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used (1)	1
01.2	<i>1 mark for each point.</i> • Age / date of the sample (radioisotope dating) (1) • Past temperatures (1) • Composition of the past atmosphere (eg CO₂ / methane concentrations) (1) • Temperature: more magnesium is incorporated into calcite at higher temperatures (1)	4
01.3	<i>1 mark for each point.</i> • Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present (1) • Ocean currents redistribute heat and nutrients; changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked (1)	2
01.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • There was no oxygen / ozone layer so high levels of UV radiation reached the land surface; water absorbed the UV and protected organisms in the oceans (1) • Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely (1) • The atmosphere contained toxic gases such as ammonia; the early Archaea were anaerobic so did not need oxygen (1) • Water acted as the solvent for reactions / transport (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	

Question	Marking guidance	Marks
02.1	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. • $600 + (90 + 150) - (220 + 80) = 540$ (1) • Stock fell by 60 so the catch exceeds the MSY / unsustainable (1)	2
02.2	1 mark for the correct answer. • $A + G - M = 90 + 150 - 80 = 160$ (1)	1
02.3	1 mark for each point. • Catch is landed and weighed at ports so it is recorded (1) • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed (1) • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years (1)	3
02.4	2 marks for each point (partial credit for a partly correct point). • CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up (2) • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
03.1	1 mark for each point. • Energy: low energy-density renewable (solar) / non-renewable fossil fuels (1) • Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion (1) • Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate (1) • Temperature: low / high (Haber process, incineration) (1)	4
03.2	1 mark for each point. • Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) (1) • They are re-used in cycles so there is no depletion of raw materials or build-up of wastes (1)	2
03.3	1 mark for each point. • Use often disperses the minerals (eg in products spread over the world / corrosion) (1) • Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced (1)	2
03.4	1 mark for each point. • They cannot easily be removed from end-of-life appliances and are not biodegradable (1) • Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> - Overall decline from 62% to 22% (1) - Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each (1)	2
04.2	<i>1 mark for each point.</i> - The polyps expel their symbiotic zooxanthellae so the coral loses colour; bleached corals are alive but grow very slowly and may die if algae do not return (1) - Raised water temperature (climate change / El Niño) (1) - Pollution: sewage, pesticides, cyanide, sunscreen; low oxygen from zooplankton blooms after overfishing (1) - High light / solar activity; ocean acidification hinders skeleton formation (1)	4
04.3	<i>1 mark for each point.</i> - Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps (1) - Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release (1) - Physical damage from anchors, divers, souvenir collection; loss of mangroves / seagrass nurseries that trap sediment (1)	3
04.4	<i>1 mark.</i> Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	
05.1	<i>1 mark for each point.</i> - The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean (1) - The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly (1)	2
05.2	<i>1 mark for each point.</i> - t-test (1) - Comparing two means of measured (continuous) data (1)	2
05.3	<i>1 mark for each point.</i> - There is a 30% probability that the difference occurred by random chance (1) - Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted (1)	2
05.4	<i>1 mark for each point.</i> - Two correlated variables may both be caused by a third variable that was not investigated (1) - Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation (1)	2
05.5	<i>1 mark for each point.</i> - Increase the number of replicates (reduces the effect of variability) (1) - Standardise other variables (soil, water, light) more closely to reduce scatter (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
06.1	<i>Correct answer only.</i> • Environmental DNA (eDNA) analysis (1)	1
06.2	<i>1 mark for each point.</i> • Transmitters are too heavy for small birds / their batteries would not last for a long-term study (1) • Geolocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded (1)	2
06.3	<i>1 mark for each point.</i> • Each separate (regional) population has its own unique gene pool (1) • DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
07.1	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early (up to 2) • Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids (up to 2) • Rotation also evens out nutrient demands and includes legumes (up to 2)	4
07.2	<i>1 mark for each point.</i> • Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) (1) • It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 (1) • Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact (1)	3
07.3	<i>1 mark for each point.</i> • Large numbers of males are sterilised with gamma radiation and released (1) • Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) (1) • Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
08.1	<i>1 mark for each point.</i> • Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change (1) • No single source could meet demand all year; battery storage on the island grid smooths fluctuations (1) • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) (1)	3
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife (1) • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles (1) • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting (1) • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection (1)	3
08.3	<i>1 mark for each point.</i> • Small groups can organise, agree and act quickly; decisions benefit people who see the results directly (1) • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away (1)	2
08.4	<i>1 mark for each point.</i> • Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand (1) • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
09.1	<i>1 mark for each point.</i> • Yield increases as inputs increase (1) • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) (1)	2
09.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Second unit: $70 - 50 = 20$ units (1) • Fifth unit: $94 - 90 = 4$ units (1)	2
09.3	<i>1 mark for each point.</i> • The first units of input on each area give the largest yield increases (1) • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve (1)	2
09.4	<i>1 mark for each point.</i> • Some areas have much greater potential productivity so investment there is sensible (1) • Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) (1)	2

Question	Marking guidance	Marks
09.5	1 mark for each point. • Yield per unit area is very high because of large inputs (1) • But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, <i>Encarsia</i>, <i>Cactoblastis</i>, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
	Model answer - Level 3 (8 marks)	
	Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, <i>Encarsia</i>, <i>Cactoblastis</i>, <i>Bacillus thuringiensis</i> - and crop wild relatives in Vavilov centres hold the genes future crops will need.	

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	Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness.	
	Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future.	
	What makes the difference	
	• Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category.	
	• Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits.	
	• 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not.	
10.2	<i>1 mark.</i> • The variety and abundance of life (species, genes, habitats) in an area / on Earth (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25